Research on Intrinsic Motivation for Reinforcement Learning

Correlation-Aware Reward Exploration (CARE)

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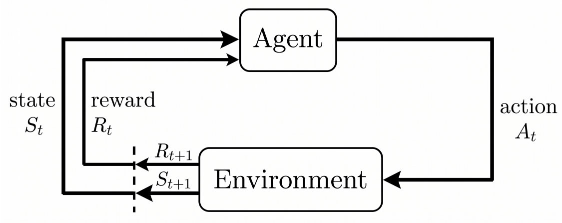
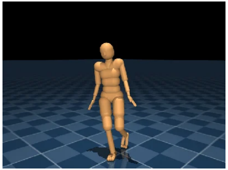
Reinforcement Learning and Sparse Reward

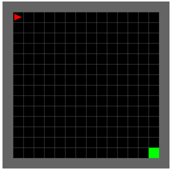


Dense Reward



MuJoCo Humanoid

Sparse Reward



MiniGrid Empty-16x16

Three Sources of Intrinsic Reward

Novelty	Surprise	Influence
"Have I been here before?"	*"Did I predict that?"*	*"Did I change the world?"*
Count-based	**ICM**	**RIDE**

Shaped reward: $r_t \;=\; r_t^{ext} + \beta\, r_t^{int}$

Fixed-β Problem and Research Question

- Same β for every state – but some states need more exploration than others.
- Same β throughout training – exploration need changes as the agent learns.
- Ignores the task – keeps exploring even when exploration does not help solve the task.

Research question

Can we adapt the intrinsic-reward coefficient automatically during training so that exploration better supports extrinsic-reward learning?

Proposed Method: CARE

Before: $r_t = r_t^{ext} + \beta \, r_t^{int}$

CARE: $r_t = r_t^{ext} + \boxed{\beta_\psi(s_t)} \, r_t^{int}$

Main idea

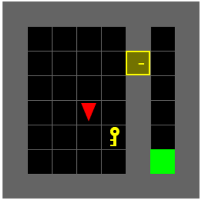
Raise intrinsic reward when exploration helps future extrinsic reward; reduce it when distracting.

CARE Training Objective

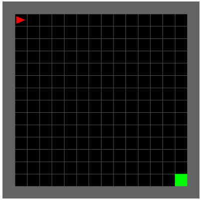
- **Correlation** – align intrinsic reward with future extrinsic advantage.
- **Regularizer** – keep β_ψ near β_0 in log-space.

$$\mathcal{L}_\beta(\psi) = \underbrace{\boxed{-\mathbb{E}\left[\,\hat{I}_z \cdot \hat{A}_z^E\,\right]}}_{\textbf{correlation}} + \underbrace{\boxed{\lambda_{\text{reg}}\left(\log \beta_\psi(s) - \log \beta_0\right)^2}}_{\textbf{log-space regularizer}}$$

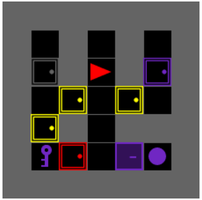
Experimental Design



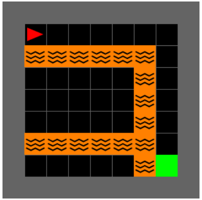
DoorKey-8x8



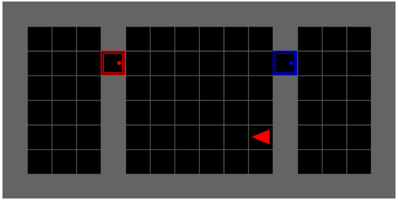
Empty-16x16



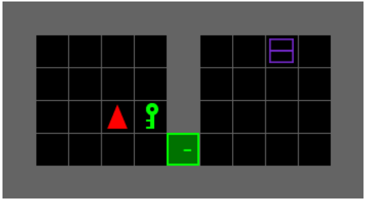
KeyCorridor-S3R3



LavaCrossing-S9N3



RedBlueDoors-8x8



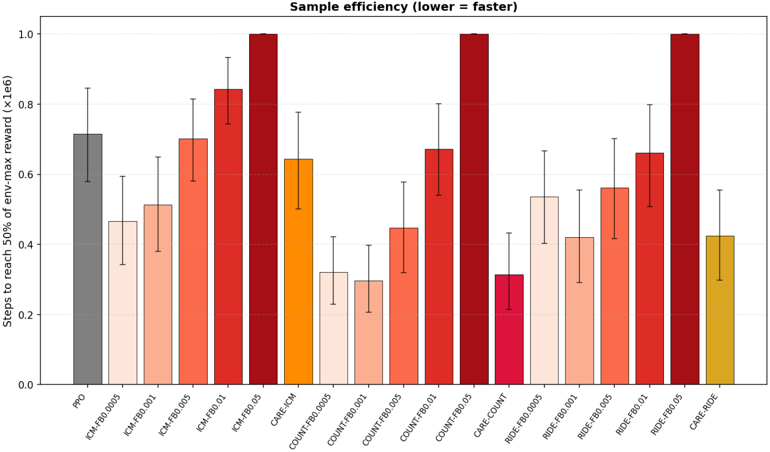
UnlockPickup

All sparse-reward, discrete grid. Red triangle = agent.

Experimental Design

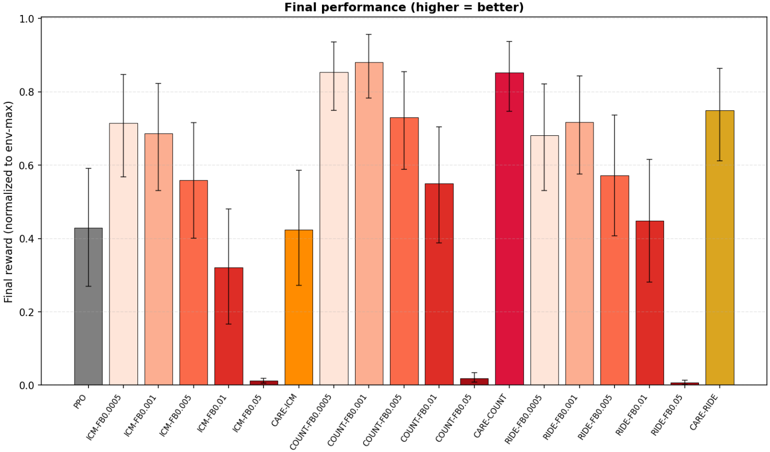
Base algorithm	PPO (actor–critic, on-policy)
Intrinsic modules	Count-based, ICM, RIDE
Compared methods	PPO (no intrinsic) / PPO + fixed-β / PPO + CARE
Environments	6 MiniGrid sparse-reward tasks (5 seeds, 10^6 env steps each)
Metrics	Success rate, episode return, learning speed, stability across seeds

Results – Sample Efficiency



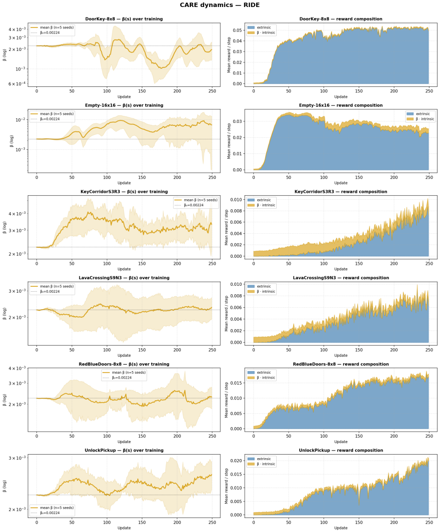
Lower is faster. Aggregated across 6 envs × 5 seeds (95% bootstrap CI).

Results – Final Performance



Higher is better. Final reward normalized to env-max, last 10 logs averaged.

Analysis – Why CARE Works



$\beta_\psi(s)$ (left) and reward composition (right) for 6 envs – CARE-RIDE.

Limitations

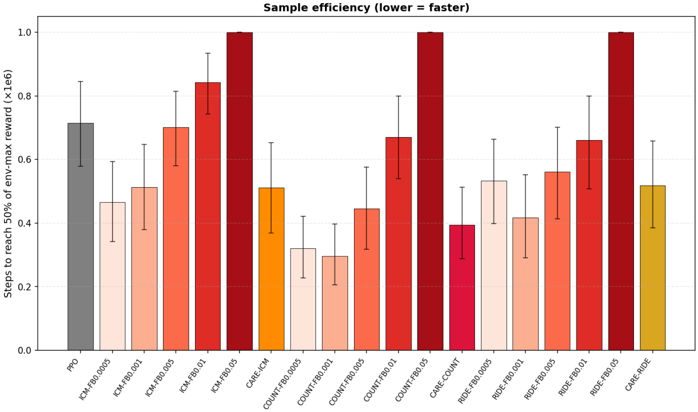
- MiniGrid only (discrete, low-dim).
- Extra network + training objective.
- Sensitive to β_0, λ_{reg}.
- Future: continuous control, image-based.

Conclusion

- Sparse rewards $\Rightarrow$ hard exploration.
- Fixed β = one-size-fits-all.
- **CARE**: adaptive $\beta_\psi(s)$ per state.

Thank you for listening!

[Backup] Extended β Range – Sample Efficiency



β sweep extended to $\beta \in [10^{-8}, 1]$.

[Backup] Extended β Range – Final Performance

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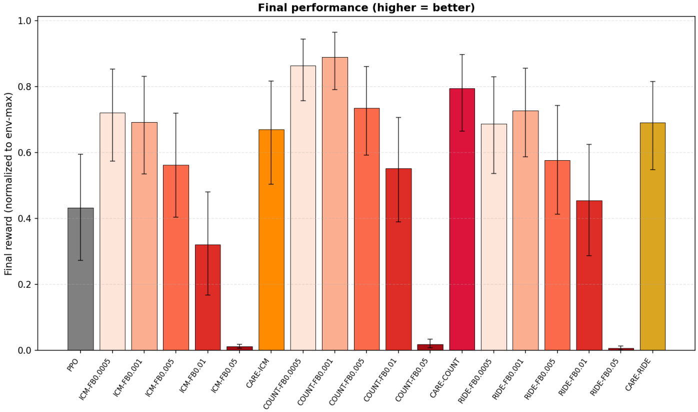

β sweep extended to $\beta \in [10^{-8}, 1]$.

[Backup] CARE vs Best Fixed β (all 18 cells)

Env	Module	CARE	Best fixed	Best β	Gap
DoorKey-8x8	COUNT	0.973	0.973	0.0005	**+0.001**
	ICM	0.770	0.973	0.0005	-0.202
	RIDE	0.973	0.973	0.0005	**+0.000**
Empty-16x16	COUNT	0.959	0.975	0.0005	-0.017
	ICM	0.575	0.975	0.0005	-0.400
	RIDE	0.941	0.974	0.001	-0.033
KeyCorridor	COUNT	0.831	0.828	0.001	**+0.003**
	ICM	0.000	0.457	0.001	-0.457
	RIDE	0.548	0.703	0.001	-0.155
LavaCrossing	COUNT	0.372	0.530	0.001	-0.158
	ICM	0.077	0.060	0.001	**+0.017**
	RIDE	0.543	0.512	0.0005	**+0.031**
RedBlueDoors	COUNT	0.653	0.779	0.0005	-0.126
	ICM	0.273	0.737	0.0005	-0.464
	RIDE	0.593	0.794	0.001	-0.201
UnlockPickup	COUNT	0.936	0.936	0.0005	**+0.000**
	ICM	0.716	0.937	0.001	-0.221
	RIDE	**0.536**	0.369	0.0005	**+0.167**

[Backup] Key Hyperparameters

PPO	$K = 80$, $\epsilon_{\text{clip}} = 0.2$, $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$ $\text{lr}_{\text{actor}} = 3\times10^{-4}$, $\text{lr}_{\text{critic}} = 10^{-3}$, rollout $T = 4{,}000$
CARE	$\beta_{\min} = 10^{-4}$, $\beta_{\max} = 5\times10^{-2}$ $\beta_0 = \sqrt{\beta_{\min}\beta_{\max}} \approx 2.24\times10^{-3}$ $\lambda_{\text{reg}} = 10^{-3}$, $\text{lr}_{\psi} = 5\times10^{-4}$, weight decay 10^{-6} one ψ-step per PPO update, grad-clip L2 $= 1.0$, no delayed target copy
Fixed-β sweep	$\beta \in \{0.0005, 0.001, 0.005, 0.01, 0.05\}$
Seeds	$\{1, 2, 3, 4, 5\}$, 10^6 env steps per run